OA-CXR: An Anatomical Retention-Guided System for Selective Chest Radiograph Report Review

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ABSTRACT

Reliable AI-assisted reporting requires identifying generated statements that warrant review. We present OA-CXR, which uses source-relative anatomical retention to prioritize chest radiograph reports. Given a presented radiograph and a categorical anatomical query, it estimates the fraction of the corresponding source-image region retained, using only the presented image and query at inference. A two-stage global–regional model learns anatomical features with auxiliary lung-segmentation supervision, then fits a standardized retention readout. A fixed parser maps eligible report statements to anatomical queries for statement-level flagging and report prioritization under fixed review budgets. Training uses controlled presentations of 19,086 source images, and evaluation covers 343,395 queries from 8,805 images across four datasets. OA-CXR achieves the lowest MAE among nine retention regressors on every dataset (macro MAE 0.0296), reducing error by 3.69% relative to TabM on shared features and 8.38% relative to a global DenseNet control trained on the same schedule. Among 6,656 MAIRA-2 reports, 445 contain eligible statements and 235 contain low-retention events. Reviewing 50% of eligible reports captures 91.7% of event-positive reports, versus 50.0% using mean token log probability. Ablations show the contributions of global–regional features and anatomical supervision. On an independent cohort of 797 pediatric radiographs, the frozen model achieves MAE 0.02810 and AUROC 0.97445 against manual clinician references. Anatomical retention thus supports prioritization of reports containing low-retention events. The source code and model-access instructions are available at `https://github.com/hekaiwang/OA-CXR`.

1. Introduction

AI-assisted reporting requires identifying generated statements that warrant review. Image acquisition and preprocessing determine the visual information available to reporting models, whose reliability varies across operating conditions [37, 44]. Changes in field of view are particularly relevant: a report may describe anatomy only partly retained in the presented image. Estimating regional retention offers an image-grounded basis for prioritizing review.

Calibration, uncertainty estimation, and failure prediction assess model outputs [10, 24, 33, 47]. Anatomical retention complements these approaches by measuring the preserved image information underlying each statement, providing a region-specific signal for review.

Chest radiograph reporting offers a practical setting. Models have advanced from memory-driven generation [6, 7] to spatially grounded descriptions and conversational assistance [3, 49], supported by clinical-entity, semantic-agreement, and factual-error evaluation measures [32, 46, 64]. We target negative statements about pleural effusion, pneumothorax, and consolidation, mapped respectively to basal, inner-rim, and whole-lung regions. These mappings connect regional retention to individual statements.

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We define *source-relative anatomical-retention estimation* as predicting the fraction of a predefined source-image region preserved in a presented radiograph. Reference lung masks and controlled crops provide continuous supervision for each query. At inference, the estimator uses only the presented image and a categorical query, yielding a region-specific measure for report review.

OA-CXR combines a two-stage global–regional retention estimator with selective report review. Stage 1 uses a DenseNet-based encoder and anatomy decoder to learn global context and regional features under joint retention and lung-segmentation supervision. Stage 2 fits a standardized readout to the frozen representations. A rule-based parser maps eligible statements to anatomical queries, converting retention scores into statement-level flags and report priorities under fixed review budgets (Fig. 1).

Following selective prediction and learning-to-defer research [14, 43], we assess retention accuracy and review allocation on 343,395 queries from 8,805 images across four datasets, alongside statement- and report-level evaluations. Shared-feature readouts, structural controls, and ablations isolate representation and readout contributions; perturbation and reduced-source studies assess presentation sensitivity and training-data dependence.

We further evaluate the frozen models on 797 pediatric radiographs from Shenzhen University of Advanced Technology General Hospital, including naturally incomplete acquisitions and 31,083 anatomical queries. Together, the four public datasets and clinical cohort cover 9,602 source images and 374,478 queries.

Our main contributions are:

1. **Source-relative anatomical-retention estimation.** We define a continuous, region-specific target for preserved source-image anatomy and learn it from controlled presentations.
2. **Anatomical-retention-guided report review.** We combine a two-stage global–regional estimator with explicit statement-to-query mappings to flag statements and prioritize reports.
3. **Evaluation of retention accuracy and review prioritization.** OA-CXR achieves the lowest MAE among nine retention regressors across four datasets. Reviewing 50% of eligible reports captures 91.7% of low-retention event-positive reports versus 50.0% using generation confidence. Ablations, perturbations, and reduced-source experiments assess model components and training choices; independent pediatric-cohort validation evaluates retention estimation and clinician-assessed report review.

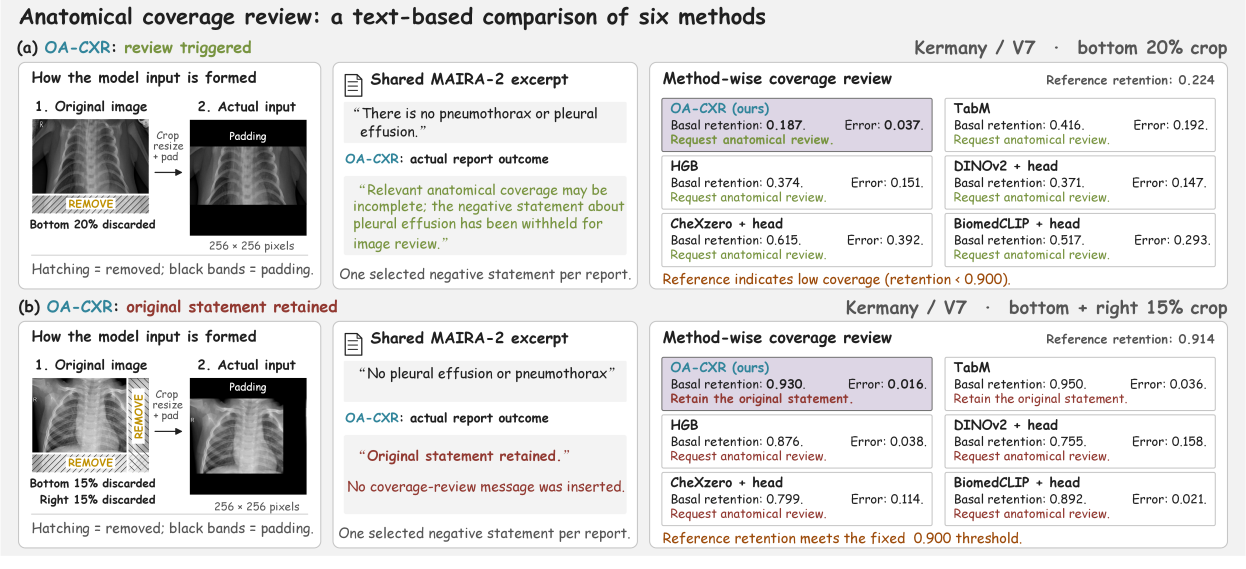


Figure 1: Retention-guided review of negative pleural-effusion statements. Each row shows the source and presented images, report excerpt, OA-CXR outcome, and basal-retention estimates from six methods. (a) The bottom 20% is removed; (b) the bottom and rightmost 15% are removed. Decisions use threshold 0.9. Hatching indicates removed regions and black bands padding; absolute errors are calculated before rounding.

2. Related Work

Report generation and grounded assistance. Early report generators used hierarchical language models [34] and retrieval–generation policies [39], followed by memory-driven transformers [7] and cross-modal memory networks [6]. MIMIC-CXR [35] pairs radiographs with reports, and Chest ImaGenome [63] adds anatomical relationships. Region-guided generation [57] and MAIRA-2 [3] ground descriptions in image regions; CheXagent [8] and RaDialog [49] extend reporting to instruction-based interpretation and conversational assistance. OA-CXR links regional retention to selected report statements in a subsequent review stage.

Global–local and anatomical representations. ConVIRT [67] learns from paired medical images and text; GLoRIA [29] combines global and local image–text features. BioViL [4] and BioViL-T [2] incorporate biomedical text semantics and temporal information, while CheXzero [58], BiomedCLIP [66], and DINOv2 [45] offer complementary pretrained features. TorchXRayVision [9] provides chest X-ray models and data interfaces. HybridGNet [16] and CheXmask [18] support anatomical segmentation. OA-CXR combines segmentation-guided regional pooling with global features to learn crop-derived retention.

Image quality and report-content assessment. AutoQC [53] and NIQE [42] measure image quality. Clinical-coherence objectives [40] and semantic rewards [13] target report quality during training. RadGraph [32], RadCliQ [64], and GREEN [46] assess report content through entities or semantic agreement, while SCUQ [60] measures factuality-related uncertainty through semantic consistency. OA-CXR instead estimates retention of predefined anatomical regions to prioritize associated statements. We compare retention-based ranking with quality and uncertainty signals for low-retention event detection.

Predictive uncertainty and failure prediction. Monte Carlo dropout [19] and deep ensembles [38] estimate predictive uncertainty, and post-hoc calibration relates confidence to correctness [24]. Distribution shifts can weaken uncertainty estimates [47]. Other methods identify misclassifications or out-of-distribution samples [26], exploit neighborhood structure [33], or learn failure-prediction confidence scores [10]. These approaches motivate scores matched to the review target. OA-CXR predicts anatomical retention and uses low-retention events as its review criterion.

Selective prediction and human–AI review. Selective prediction balances prediction coverage against the risk of retained cases [14, 20]; SelectiveNet [21] jointly learns prediction and rejection. Learning-to-defer methods incorporate expert decisions or optimize referral [41, 43], while complementary learning addresses joint human–model performance [62]. Medical second-opinion systems can learn target-specific uncertainty scores [52]; conformal risk control [1] provides statistical guarantees for selection. OA-CXR ranks reports by predicted anatomical retention and assesses empirical risk–coverage and fixed-budget event recall.

Presentation sensitivity and data dependence. Common-corruption benchmarks [25] and WILDS [37] examine perturbations and real distribution shifts. Shortcut learning [22], underspecification [11], and hidden stratification [44] motivate evaluation beyond aggregate scores. Chest-radiograph studies document source-dependent generalization [65] and nonpathological cues in COVID-19 prediction [12]. We test cropping, appearance, padding, and training-source composition; Section 4.1 details the image and annotation sources.

3. Method

Figure 2 summarizes the workflow; Fig. 3 details the architecture. Given a presented radiograph x and query $q \in \mathcal{Q} = \{\text{basal}, \text{rim}, \text{whole}\}$, OA-CXR estimates source-relative retention of the corresponding region. Stage 1 jointly learns retention and anatomical representations with segmentation supervision; Stage 2 fits the final readout to frozen features. A separate parser maps report statements to anatomical queries.

3.1. Mask-derived Retention Supervision

Let I denote a source radiograph, M_l the reference mask for lung $l \in \{1, 2\}$, and C a crop window in source-image coordinates. For each lung, we define three reference regions $R_{lq}(I)$: the whole-lung mask; a basal region covering the inferior quarter of its source-image bounding box; and an inner-rim region given by an interior Euclidean-distance band of radius $\max(1, 0.02s_l)$, where s_l is the shorter bounding-box dimension. Fractional boundaries use continuous pixel-area integration.

The retention target is

$$y_q(I, C) = \min_{l \in \{1,2\}} \frac{|R_{lq}(I) \cap C|}{|R_{lq}(I)|}. \tag{1}$$

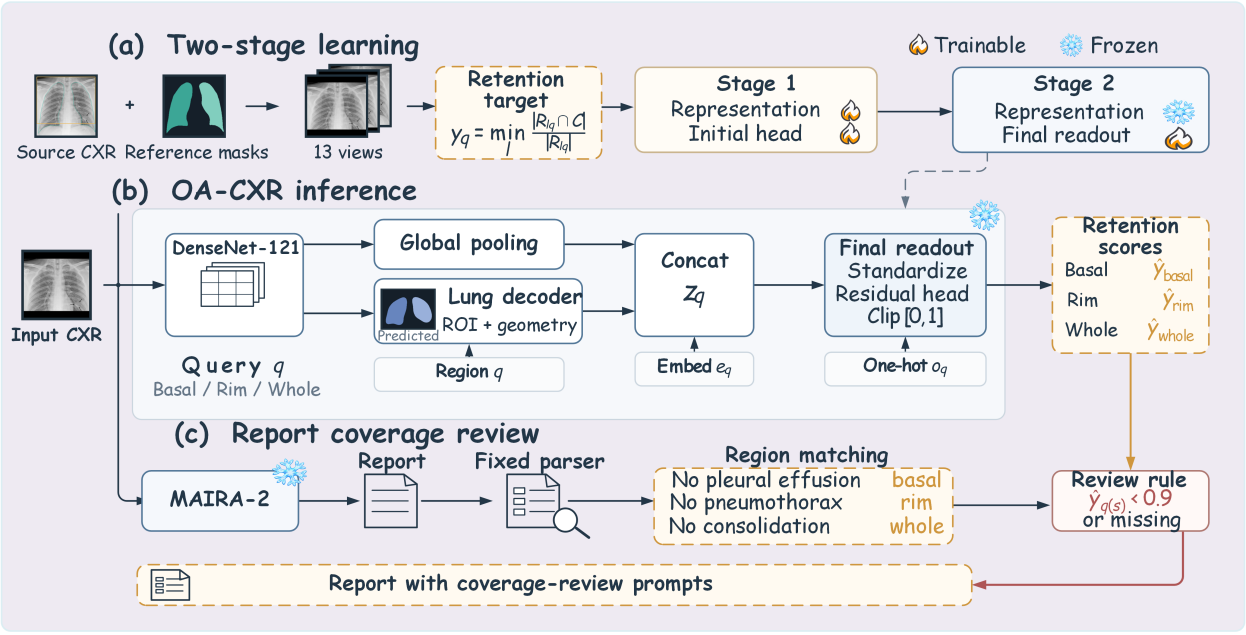


Figure 2: OA-CXR workflow. (a) Source radiographs and lung masks define retention targets for thirteen controlled presentations. Stage 1 learns anatomical features and an initial retention head; Stage 2 freezes the representation and fits the final readout. (b) At inference, predicted anatomical regions and the specified query yield a retention score from the presented image, without source images or reference masks. (c) A fixed parser links eligible negative MAIRA-2 statements to anatomical queries; low or unavailable scores trigger coverage-review notices.

The minimum captures anatomical loss on either side. Targets are computed before resizing and padding, in source-image coordinates. Uncropped images have retention one, and transformations preserving anatomical support leave the target unchanged.

Source-image masks define the training targets in Eq. 1; at inference, OA-CXR predicts them from learned global and regional features of the presented image.

Each source image generates thirteen presentations: the original, eight single-edge crops removing 10% or 20%, and four corner crops removing 15% from two adjacent edges. Images are aspect-ratio-preserving resized and zero-padded to 256×256. Bilinear interpolation is used for intensities and nearest-neighbor interpolation for masks; intensities are mapped to $[-1024, 1024]$. Inference uses only the resulting image and query.

3.2. Global–Regional Anatomical Representation

A DenseNet-121 encoder [28], initialized with CheXpert-trained TorchXRayVision weights [9, 30], maps a $1 \times 256 \times 256$ image to $F \in \mathbb{R}^{1024\times8\times8}$; global average pooling produces $g \in \mathbb{R}^{1024}$. The anatomy decoder projects F to 64 channels, then applies two bilinear upsampling blocks with 3×3 convolutions, four-group normalization, and SiLU, producing $A \in \mathbb{R}^{64\times32\times32}$. A separate 1×1 convolution and sigmoid yield two lung probability maps $P \in [0, 1]^{2\times32\times32}$.

Regional supports use predicted lung probabilities: $P_l(p) \geq 0.5$ defines membership, while probabilities serve as pooling weights. Basal support occupies the inferior quarter of the predicted lung bounding box. Inner-rim support uses square-kernel erosion with radius $\max(1, \lceil 0.02\widetilde{s}_l \rceil)$, where $\widetilde{s}_l$ is the shorter predicted bounding-box dimension on the decoder grid. These pooling supports are distinct from the source-image reference regions defining supervision in Eq. 1.

Let W_{lq} be the predicted weight map for lung l and query q, with $W_q(p) = \max_l W_{lq}(p)$. On decoder grid Ω, weighted pooling is

$$\Pi(A, W) = \frac{\sum_{p\in\Omega} W(p)A(:, p)}{\max\left(\sum_{p\in\Omega} W(p), \epsilon\right)}, \qquad \epsilon = 10^{-6}. \tag{2}$$

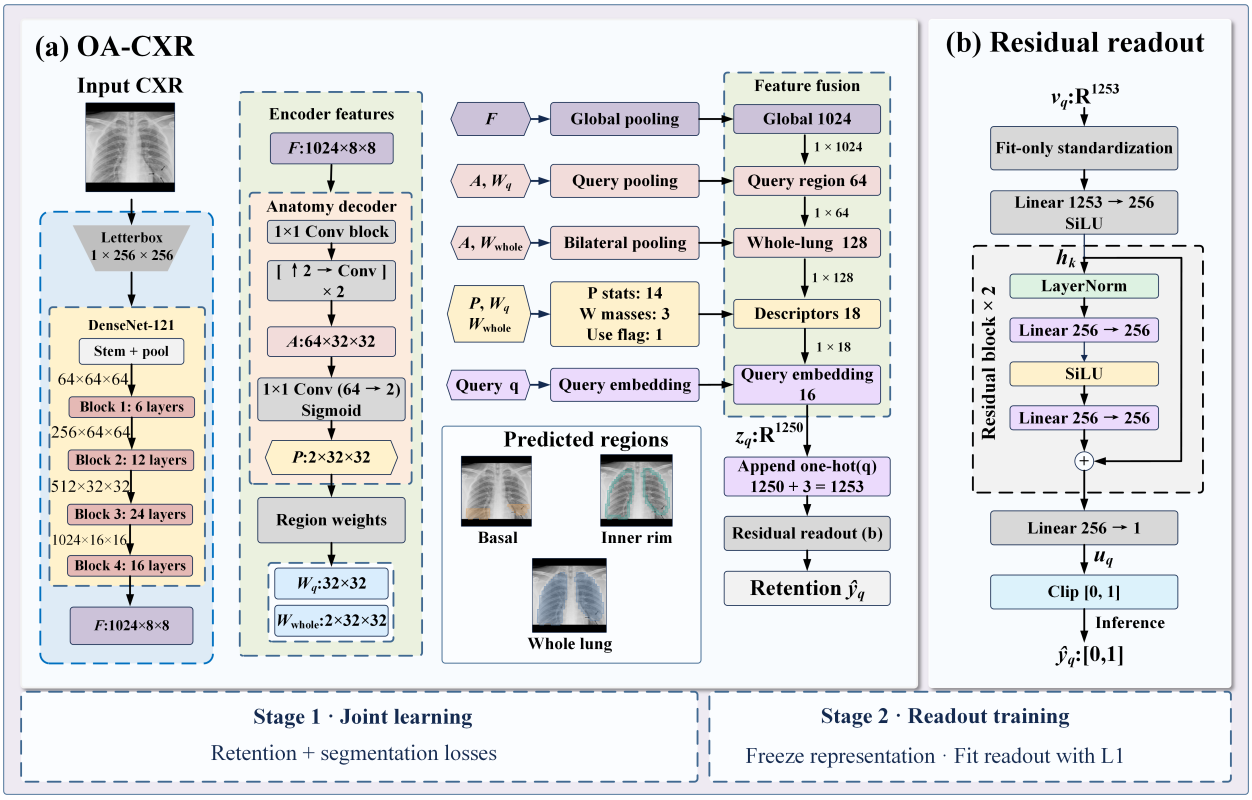


Figure 3: OA-CXR architecture. (a) DenseNet-121 provides global features; the anatomy decoder predicts lung probabilities for query-specific and bilateral pooling. Global, regional, geometric, and query features form a 1,250-dimensional representation; a three-dimensional one-hot query code gives 1,253 readout inputs. (b) Fit-only standardization precedes a 256-dimensional projection, two residual blocks, and a scalar output clipped to $[0, 1]$ for scoring. Stage 1 uses an auxiliary retention head; Stage 2 trains the residual readout with L1 loss on frozen representations.

The query-region vector is $r_q = \Pi(A, W_q) \in \mathbb{R}^{64}$, and bilateral whole-lung vectors are $b_l = \Pi(A, W_{l,\text{whole}}) \in \mathbb{R}^{64}$. Empty supports produce zero vectors.

The representation concatenates global, regional, bilateral, geometric, and query features:

$$z_q = [g; r_q; b_1; b_2; d_q; e_q] \in \mathbb{R}^{1250}, \tag{3}$$

Here $d_q \in \mathbb{R}^{18}$ contains 14 probability-map statistics, three normalized support masses, and a regional-branch-use flag (Fig. 3); $e_q \in \mathbb{R}^{16}$ is a learned query embedding. The dimensions sum to $1024 + 64 + 128 + 18 + 16 = 1250$.

3.3. Stage 1: Joint Representation Learning

An initial shared head maps z_q through $1250 \to 128 \to 1$ with SiLU and sigmoid output. The encoder, anatomy decoder, lung head, embedding, and retention head jointly minimize

$$\mathcal{L}_{\text{repr}} = \mathcal{L}_{\text{ret}} + 0.25\left(\mathcal{L}_{\text{BCE}} + 0.5\mathcal{L}_{\text{Dice}}\right), \tag{4}$$

where $\mathcal{L}_{\text{ret}}$ is Smooth L1 loss ($\beta = 0.1$), averaged across queries. Segmentation losses compare transformed reference masks with upsampled lung predictions.

During training, the query-region vector, bilateral context, and related geometric descriptors are jointly dropped with probability 0.2; global features, query embeddings, and segmentation supervision remain active. Encoder BatchNorm running statistics are fixed, but convolutional weights and affine parameters remain trainable. The initial retention head is discarded after Stage 1 checkpoint selection.

3.4. Stage 2: Standardized Residual Readout

In Stage 2, representation components are frozen and regional dropout is disabled. We append a three-dimensional one-hot query code o_q to Eq. 3:

$$v_q = [z_q; o_q] \in \mathbb{R}^{1253}, \qquad \widetilde{v}_q = (v_q - \mu_{\text{fit}}) \oslash s_{\text{fit}}. \tag{5}$$

Featurewise means and population standard deviations are fitted on training features and fixed thereafter; values below 10^{-6} are replaced by one.

A linear projection and SiLU map standardized inputs to $h_0 \in \mathbb{R}^{256}$ (Fig. 3(b)); two residual blocks compute

$$h_{k+1} = h_k + W_{k2}\,\text{SiLU}\big(W_{k1}\,\text{LN}(h_k) + b_{k1}\big) + b_{k2}, \qquad k \in \{0, 1\}, \tag{6}$$

followed by the scalar output

$$u_q = w_{\text{out}}^{\top} h_2 + b_{\text{out}}. \tag{7}$$

Only the readout parameters minimize

$$\mathcal{L}_{\text{readout}} = \frac{1}{|\mathcal{D}_{\text{fit}}||\mathcal{Q}|} \sum_{x \in \mathcal{D}_{\text{fit}}} \sum_{q \in \mathcal{Q}} |u_q(x) - y_q(x)|. \tag{8}$$

Here $\mathcal{D}_{\text{fit}}$ contains the presented fit images. The reported prediction is

$$\hat{y}_q = \min(1, \max(0, u_q)). \tag{9}$$

Training uses unclipped outputs; scoring clips to [0, 1]. Development MAE selects the checkpoint. One encoder–decoder pass serves all three queries.

3.5. Report Coverage Review

Frozen MAIRA-2 [3] generates reports. A fixed rule-based parser maps explicit negative statements about pleural effusion, pneumothorax, and consolidation to basal, inner-rim, and whole-lung queries, respectively. For eligible statement s,

$$e_s = \mathbf{1}\big[y_{q(s)} < \tau_{\text{event}}\big], \qquad \rho_s = 1 - \hat{y}_{q(s)}, \tag{10}$$

with $\tau_{\text{event}} = 0.9$. Statements are flagged if $\hat{y}_{q(s)} < \tau_{\text{review}} = 0.9$ or no score is available. The first threshold defines reference events; the second governs review. A deterministic policy withholds or restructures flagged statements and adds a coverage-review notice. Evaluation covers parser-eligible statements and their reports.

For a report d with eligible statements S_d,

$$\rho_d = \max_{s \in S_d} \rho_s, \qquad e_d = \max_{s \in S_d} e_s. \tag{11}$$

Eligible reports are ranked by decreasing ρ_d with review budgets of 5%, 10%, 20%, 30%, and 50%. Ties at budget boundaries receive equal fractional selection probabilities. Unavailable scores mandate review; none occur in the fixed-workload comparison.

3.6. Independent Retention–Confidence Combination

We test whether generation confidence improves retention-based ranking with a logistic combination fitted on a separate calibration panel. For statement s in report d, let $a_s = 1 - \hat{y}_{q(s)}$ and $\ell_d = -\overline{\log p}_d$, where $\overline{\log p}_d$ is teacher-forced mean token log probability. The combined score is

$$\rho_s^{\text{comb}} = \sigma\left(b + \beta_1 \frac{a_s - \mu_1}{s_1} + \beta_2 \frac{\ell_d - \mu_2}{s_2}\right). \tag{12}$$

The model fits low-retention events e_s, weighting statements inversely by eligible-statement counts per source group to equalize group weight. Weighted means and population standard deviations use calibration data only. Logistic fitting uses L2 regularization ($C = 1$), LBFGS, at most 1,000 iterations, tolerance 10^{-4}, and seed 17. Parameters and scaling statistics remain fixed at evaluation.

A whole-lung-only control assigns whole-lung scores to all eligible statements while retaining finding-specific event labels, isolating the effect of query-specific review ranking.

Table 1
Dataset partitions and standard evaluation sizes after duplicate screening. Source-image counts precede thirteen presentations and three queries per presentation. Resampling groups are Kermany/V7 filename families, clinical-cohort identity groups, and individual source images for other datasets (Section 3.1).

	Source images				Standard evaluation		
Source	Fit	Dev.	Calib.	Evaluation	Presentations	Queries	Groups
COVID-QU-Ex	16,192	2,255	2,289	1,855	24,115	72,345	1,855
Kermany/V7	2,894	354	362	937	12,181	36,543	360
NIH/CheXmask	0	0	0	5,821	75,673	227,019	5,821
Shenzhen	0	0	0	192	2,496	7,488	192
Clinical cohort	0	0	0	797	10,361	31,083	791
Total	19,086	2,609	2,651	9,602	124,826	374,478	9,019

4. Experiments

4.1. Datasets

We evaluate OA-CXR on five chest-radiograph cohorts: COVID-QU-Ex [55, 56], Kermany/V7 [36, 59], NIH/CheXmask [18, 61], Shenzhen [31, 54], and an external pediatric cohort from Shenzhen University of Advanced Technology General Hospital. COVID-QU-Ex and Kermany/V7 supply fitting and checkpoint-selection data; NIH/CheXmask, Shenzhen, and the clinical cohort are evaluated only. Lung masks define reference regions for Eq. 1. COVID-QU-Ex uses human–machine collaborative annotations [56]; CheXmask provides HybridGNet-derived masks [16, 18] via PhysioNet [17, 50]; clinicians annotated the clinical cohort.

Exact and perceptual image fingerprints identified 4,541 duplicate source images, which were excluded from the public cohorts. COVID-QU-Ex was divided by deterministic hashing into fit, development, calibration, and held-out evaluation sets. Kermany images were partitioned by filename family, with families containing official test images assigned to evaluation. All presentations from a source image remained in one partition.

The clinical cohort comprised 800 pediatric radiographs, mainly from respiratory-infection-related cases; removing three exact duplicates left 797 images. Ages ranged from 10 days to 13 years (median, 3 years 7 months). The images formed 791 identity groups for resampling and were used only for external evaluation. Clinicians annotated bilateral lungs and assessed report-review outcomes. The retrospective study was approved by the Institutional Review Board (IRB)/Ethics Committee of Shenzhen University of Advanced Technology General Hospital (approval no. LL-KT-2026118; September 28, 2026), which waived informed consent for de-identified retrospective data.

Each image followed the thirteen-presentation protocol (Section 3.1), with basal, inner-rim, and whole-lung queries for each presentation. Table 1 gives image, presentation, query, and resampling-group counts. The clinical cohort contributed 10,361 presentations and 31,083 queries, including 8,589 (27.63%) with reference retention below 0.9.

4.2. Implementation and Comparison Methods

OA-CXR uses Python 3.10, PyTorch 2.6, and TorchXRayVision 1.5.4 for encoder initialization. Both stages use AdamW and seed 17. Stage 1 trains for 13 epochs (batch size 256; learning rate 10^{-4}; weight decay 10^{-4}; BF16; regional dropout 0.2). A deterministic cyclic schedule covers all 13 presentations per source image in 975 optimizer updates; development performance selects epoch 12. Stage 2 trains the frozen-feature readout for 30 epochs over all queries (batch size 2,048; FP32; weight decay 10^{-4}; cosine learning rate from 10^{-3} to 10^{-5}). Development MAE selects the L1 readout at epoch 27.

Eight regression comparators include five readouts fitted to the same frozen OA features—histogram gradient boosting (HGB) [15, 48], XGBoost [5], CatBoost [51], Ridge [27], and TabM [23]—and three frozen pretrained encoders with fitted readouts: BiomedCLIP [66], CheXzero [58], and DINOv2 [45]. Additional image-based ranking signals are CheXpert-only [30], AutoQC CXR-NIQE [42, 53], and BioViL-T [2]. Report-level comparators include mean token log probability, sentence-level log probability, negative token entropy, and SCUQ-RadGraph [32, 60]. Quality, confidence, and uncertainty scores are compared by ranking metrics.

Table 2
Anatomical-retention regression on four evaluation sources. Columns give source-specific MAE and equally weighted macro MAE/RMSE. Lower is better; bold marks the best result.

Method	COVID-QU-Ex	Kermany/V7	NIH/CheXmask	Shenzhen	Macro MAE	Macro RMSE
HGB	0.03633	0.04708	0.03350	0.03727	0.03855	0.07089
XGBoost	0.03828	0.04878	0.03548	0.03873	0.04032	0.07313
CatBoost	0.03815	0.04909	0.03521	0.03892	0.04034	0.07289
Ridge	0.04421	0.05266	0.04110	0.04326	0.04531	0.07665
TabM	0.02823	0.03748	0.02655	0.03080	0.03077	0.06475
BiomedCLIP + head	0.04910	0.05454	0.04673	0.04522	0.04890	0.09341
CheXzero + head	0.03988	0.04805	0.03728	0.04275	0.04199	0.08414
DINOv2 + head	0.03881	0.04758	0.03634	0.03293	0.03892	0.07634
OA-CXR (ours)	**0.02695**	**0.03683**	**0.02534**	**0.02940**	**0.02963**	**0.06265**

Table 3
Statement-level low-retention detection and selective review: AUROC↑ / AP↑ / AURC↓. Methods share valid statements within each panel; bold marks the best point estimate.

(a) Comparison including report uncertainty: 741 common statements			
Method	COVID-QU-Ex ($n = 136$)	Kermany/V7 ($n = 432$)	NIH/CheXmask ($n = 173$)
HGB	0.9866 / 0.9787 / 0.0949	0.9737 / 0.9731 / **0.1788**	0.9825 / 0.9827 / 0.1615
XGBoost	0.9866 / 0.9792 / 0.0949	0.9715 / 0.9697 / 0.1798	0.9755 / 0.9756 / 0.1648
CatBoost	0.9836 / 0.9756 / 0.0962	0.9718 / 0.9646 / 0.1792	0.9751 / 0.9743 / 0.1647
Ridge	0.9834 / 0.9759 / 0.0964	0.9663 / 0.9594 / 0.1823	0.9501 / 0.9472 / 0.1836
CheXpert-only	0.4635 / 0.3607 / 0.3763	0.4716 / 0.5322 / 0.5631	0.4960 / 0.5320 / 0.4916
AutoQC CXR-NIQE	0.5333 / 0.4607 / 0.3640	0.6081 / 0.5979 / 0.4521	0.5112 / 0.4904 / 0.4872
BioViL-T	0.5483 / 0.4597 / 0.3423	0.4996 / 0.5124 / 0.5293	0.4559 / 0.4595 / 0.5048
Report log probability	0.4544 / 0.3690 / 0.4516	0.5386 / 0.5921 / 0.5257	0.3565 / 0.4185 / 0.6359
Sentence log probability	0.3867 / 0.3587 / 0.5119	0.5179 / 0.5606 / 0.5548	0.5210 / 0.5193 / 0.4627
Negative token entropy	0.3826 / 0.3385 / 0.4913	0.5040 / 0.5499 / 0.5645	0.5321 / 0.5226 / 0.4452
SCUQ-RadGraph	0.4732 / 0.3832 / 0.4151	0.5406 / 0.5451 / 0.4897	0.3999 / 0.4469 / 0.5864
OA-CXR (ours)	**0.9952** / **0.9915** / **0.0915**	**0.9810** / **0.9846** / 0.1797	**0.9921** / **0.9927** / **0.1570**
(b) Comparison with additional regressors: 784 common statements			
Method	COVID-QU-Ex ($n = 141$)	Kermany/V7 ($n = 464$)	NIH/CheXmask ($n = 179$)
TabM	0.9889 / 0.9820 / 0.1014	0.9800 / 0.9801 / **0.1703**	0.9899 / 0.9888 / 0.1463
DINOv2 + head	0.9524 / 0.9499 / 0.1288	0.9673 / 0.9647 / 0.1785	0.9729 / 0.9722 / 0.1549
CheXzero + head	0.9519 / 0.9403 / 0.1226	0.9658 / 0.9663 / 0.1772	0.9527 / 0.9104 / 0.1613
BiomedCLIP + head	0.9263 / 0.8771 / 0.1278	0.9403 / 0.9394 / 0.1904	0.9577 / 0.9472 / 0.1604
OA-CXR (ours)	**0.9912** / **0.9869** / **0.1006**	**0.9803** / **0.9830** / 0.1742	**0.9926** / **0.9927** / **0.1453**

Two structural controls isolate anatomical modeling and learned image features. Global DenseNet uses the same initialization and training schedule without the anatomy decoder, segmentation loss, or regional pooling. The predicted-mask geometry control trains HGB on 75 geometric and query features from the frozen OA segmentation model.

Clinical evaluation uses frozen models without cohort-specific training, checkpoint selection, or score-direction adjustment. The 14 configurations comprise eight regression comparators, three image-based ranking signals, two structural controls, and OA-CXR. MAE and RMSE are reported for 11 retention regressors; ranking metrics cover all 14 configurations.

4.3. Metrics and Statistical Analysis

Retention regression uses mean absolute error (MAE) and root mean squared error (RMSE). Macro scores equally average the four source-specific metrics. For low-retention events ($y_q < 0.9$), we report AUROC, average precision (AP), and area under the risk–coverage curve (AURC), where risk is the low-retention proportion among retained cases and

Table 4
Fixed-threshold review outcomes for 784 eligible statements ($\hat{y}_q < 0.9$). "Low-retention unflagged" means $y_q < 0.9$ and $\hat{y}_q \geq 0.9$; "Above-threshold flagged" means $y_q \geq 0.9$ and $\hat{y}_q < 0.9$. Lower error counts are better; total flags indicate workload.

Method	Low-retention unflagged	Above-threshold flagged	Total flagged
TabM	35	**17**	361
DINOv2 + head	35	37	381
CheXzero + head	43	38	374
BiomedCLIP + head	50	60	389
OA-CXR (ours)	**29**	**17**	367

coverage is their retained fraction. Fixed-budget review uses event recall and the event proportion among unreviewed eligible reports. Comparisons use common valid cases; missing scores and denominators are reported separately.

We compute 95% confidence intervals using 1,000 paired grouped bootstrap resamples, keeping all presentations and queries within each group. Resampling units are Kermany filename families or source images for other datasets (Table 1); all methods share resamples. Pooled workload analyses are also stratified by dataset. With all models fitted using seed 17, intervals reflect evaluation-sample uncertainty conditional on the fitted models. Ablation intervals are unadjusted.

Clinical-cohort metrics are separate from four-source macro averages. Pointwise 95% percentile intervals use 1,000 paired bootstrap resamples of 791 identity groups, retaining all images, presentations, and queries in each group. Methods share resamples, and paired between-method intervals have no multiplicity adjustment.

For Fig. 6(e–l), we use approximate two-sided paired delete-one-group jackknife t-tests (790 degrees of freedom), with Holm correction for 80 comparisons: ten comparators against OA-CXR on overall MAE, AUROC, and three region-specific MAEs and RMSEs. Confidence intervals use grouped bootstrap resampling; significance annotations use Holm-adjusted jackknife tests.

4.4. Comparison with Other Methods

Across 343,395 public-dataset queries, OA-CXR has the lowest MAE point estimate among nine retention regressors on all four datasets (Table 2). Its macro MAE is 0.02963 versus 0.03077 for TabM and 0.03855 for HGB, reductions of 3.69% and 23.13% from unrounded values. OA-CXR also has the lowest source-specific RMSE point estimates. Figure 4 combines prediction agreement, low-retention ranking, risk–coverage, and error uncertainty.

The histograms in Fig. 4(a) compare predictions with reference retention; values above and below the identity line indicate over- and underestimation. OA-CXR clusters most closely around this line, consistent with its lowest macro MAE. Normalization within datasets gives all four sources equal weight despite different query counts. TabM is the closest regression comparator on the same frozen OA features, indicating the readout contribution. Among alternative frozen encoders, DINOv2 has the lowest macro MAE (0.03892), followed by CheXzero and BiomedCLIP.

The paired grouped-bootstrap distributions in Fig. 4(d) show source-specific MAE and RMSE variability. Paired MAE intervals favor OA-CXR over TabM on all four datasets, complementing the prediction agreement in panel (a).

For the low-retention event $y_q < 0.9$, OA-CXR achieves the highest AUROC point estimate on all four datasets (0.9705–0.9836) and the highest AP on COVID-QU-Ex, Kermany/V7, and NIH/CheXmask (Fig. 4(b)). Adapted CheXpert-only, AutoQC, and BioViL-T signals discriminate less effectively (AUROC 0.4170–0.5462). Risk–coverage curves (Fig. 4(c)) show that withholding lower-retention queries reduces low-retention prevalence among retained queries, which rises as coverage approaches 100%. The figure thus links retention accuracy, event detection, and selective query retention.

Figure 1 shows two review decisions. In panel (a), basal retention is 0.224 and OA-CXR predicts 0.187, the smallest absolute error among six methods; all request review. In panel (b), reference retention is 0.914 and OA-CXR predicts 0.930. OA-CXR and TabM retain the statement, while four other methods flag it. The cases illustrate decisions below and near the 0.9 threshold.

The 6,656 generated reports contain 784 parser-eligible negative statements. Table 3 uses common valid statements: 741 in panel (a), after excluding 43 missing SCUQ scores, and all 784 in panel (b). Shenzhen has no eligible statements, so statement-level metrics are undefined there.

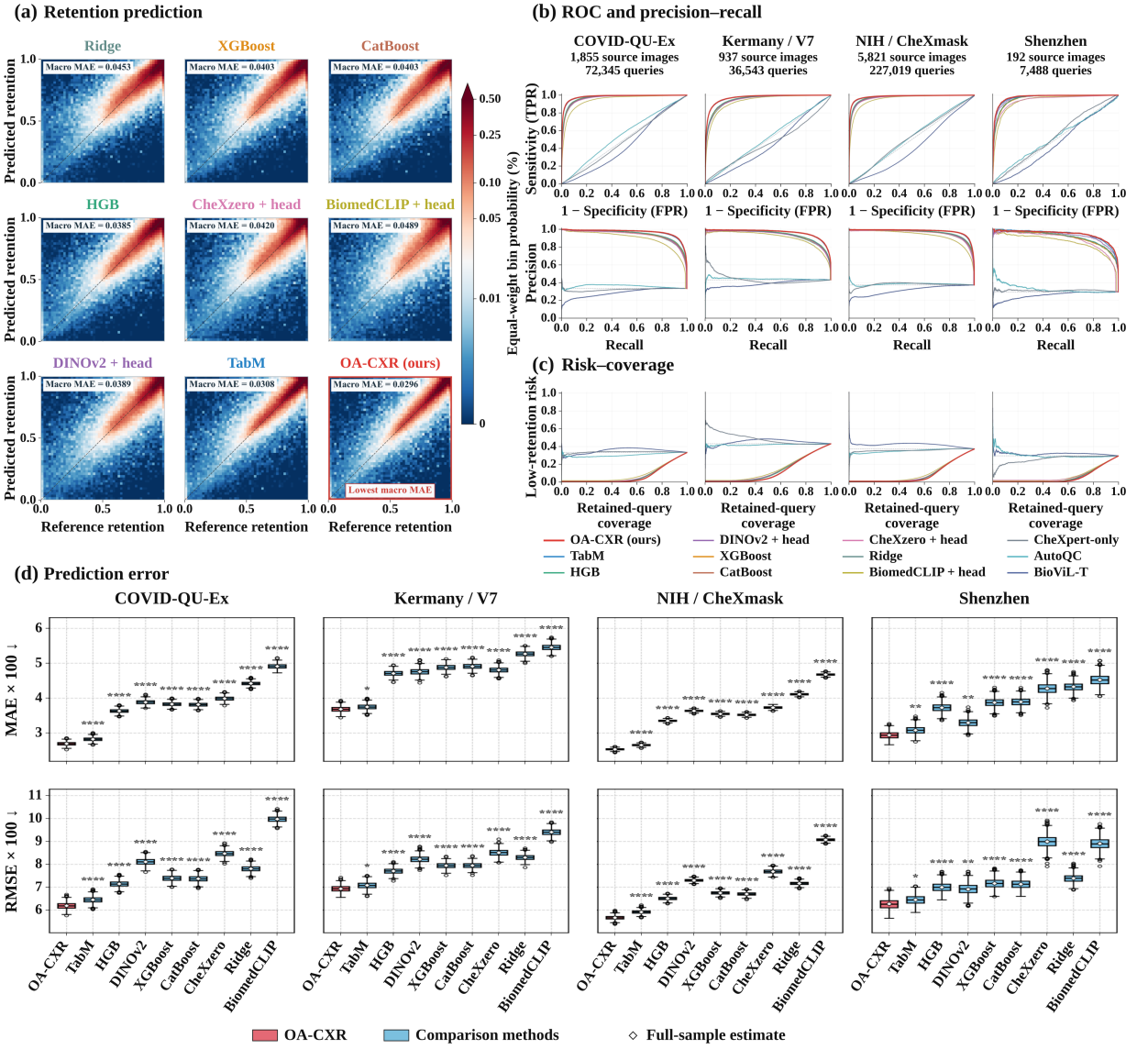


Figure 4: Public-dataset retention estimation and low-retention event detection (8,805 source images; 343,395 queries). **(a)** Predicted versus reference retention for nine regressors. Each 50×50 histogram is normalized to probabilities within its dataset, then averaged across datasets (25% each). The shared log color scale saturates at 0.50% per bin; dashed lines mark exact agreement and annotations show macro MAE. **(b)** ROC (top) and precision–recall (bottom) curves for twelve image-based methods, by dataset, for $y_q < 0.9$. **(c)** Risk–coverage curves for the same methods, admitting queries in decreasing order of predicted retention or other higher-is-better scores. Risk is low-retention prevalence among retained queries. Higher curves are preferable in (b), lower in (c). **(d)** Source-specific MAE (top) and RMSE (bottom), multiplied by 100, for nine regressors across 1,000 paired grouped-bootstrap resamples that preserve grouped queries and share resamples across methods. Boxes show interquartile ranges; lines, medians; whiskers, 1.5×IQR; circles, outliers; diamonds, full-sample estimates. Adjusted significance: $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$; ns, $P \geq 0.05$.

Table 5
Fixed-representation and full-pipeline ablations. Dev. indicates development-selected readouts; Last, their epoch-30 counterparts. Percentage increases use unrounded values relative to the corresponding Full control. Paired 95% intervals show $\mathrm{MAE}_{\mathrm{Full}} - \mathrm{MAE}_{\mathrm{variant}}$ at Dev. in 10^{-3} units; negative favors Full. Bold marks the lowest macro MAE within each panel.

Condition	Dev. macro MAE	Increase (%)	Last macro MAE	95% CI ($\times 10^{-3}$)
(a) Fixed-representation and readout ablations				
Without regional inputs	0.030893	+3.67	0.030893	[−1.3312, −0.8522]
Without global inputs	0.031654	+6.23	0.031698	[−2.1964, −1.5587]
Without geometric descriptors	0.029935	+0.46	0.030102	[−0.2942, +0.0228]
Without explicit query encoding	0.029886	+0.29	0.030069	[−0.2644, +0.0878]
Without standardization	0.031732	+6.49	0.031732	[−2.3128, −1.5914]
Without residual connections	0.029843	+0.15	0.029880	[−0.3264, +0.2186]
Smooth L1 instead of L1	0.030064	+0.89	0.030358	[−0.4265, −0.0981]
Sigmoid output	0.029921	+0.41	0.029975	[−0.3755, +0.1571]
OA-CXR (Full)	**0.029799**	—	**0.029867**	—
(b) Full-pipeline ablations				
Without segmentation loss	0.032257	+8.15	0.032434	[−2.8432, −2.0532]
Without regional features	0.030481	+2.19	0.030638	[−0.9782, −0.3446]
Without regional-branch dropout	0.030068	+0.81	0.030068	[−0.4872, −0.0033]
Frozen image encoder	0.036316	+21.76	0.036405	[−7.0562, −5.9522]
OA-CXR (Full)	**0.029827**	—	**0.029827**	—

OA-CXR has the highest statement-level AUROC and AP point estimates on all three evaluable datasets in both panels, and the lowest AURC on COVID-QU-Ex and NIH/CheXmask. Report-probability and uncertainty scores perform worse on AUROC and AP for this target.

At threshold $\hat{y}_q < 0.9$, OA-CXR leaves 29 low-retention statements unflagged versus 35 for TabM; both flag 17 statements with reference retention at least 0.9 (Table 4). Frozen-encoder methods miss 35–50 low-retention statements and flag 37–60 above-threshold statements. OA-CXR has the fewest missed low-retention statements among these five methods and ties TabM for the fewest above-threshold flags.

4.5. Ablation Studies

We evaluate nine fixed-representation conditions, retraining only the readout, and five full-pipeline conditions, retraining both stages, each with a separate Full control. All use the same partitions and seed 17, with 30 readout epochs and, for pipeline variants, 13 image-training epochs. Table 5 reports development-selected readouts (Dev.) and epoch-30 counterparts (Last) from development-selected image checkpoints. The respective Full controls yield macro MAEs of 0.029799 and 0.029827. Figure 5(a,b) shows the bootstrap distributions, panel (c) quantifies component-wise error increases, panel (d) compares full-pipeline accuracy, and panel (e) summarizes statement-review errors.

Removing regional inputs, global inputs, or standardization increases fixed-representation macro MAE by 3.67%, 6.23%, or 6.49%, respectively; replacing L1 with Smooth L1 increases it by 0.89% (Fig. 5(a,c)). Paired Dev.-endpoint intervals exclude zero for these four comparisons. Geometric descriptors, explicit query encoding, residual connections, and output activation have smaller effects (0.15–0.46%; Table 5(a)). Query-dependent regional pooling remains in the variant without explicit query encoding.

In full-pipeline ablations, freezing the encoder increases macro MAE by 21.76%, removing segmentation supervision by 8.15%, and removing regional features by 2.19% (Fig. 5(b–d)). Removing regional-branch dropout increases MAE by 0.81%. Paired Dev.-endpoint intervals exclude zero for all four comparisons (Table 5(b)). Full has the lowest macro MAE in both ablation families.

Segmentation supervision also improves localization and statement review. Removing it reduces Dice from 0.908–0.931 for Full to 0.178–0.231 (Table 6(a)). At the 0.9 review threshold, the full-pipeline Full control leaves 29 low-retention statements unflagged and flags 14 above-threshold statements, versus 33–39 and 18–22, respectively, for the four ablations (Fig. 5(e); Table 6(b)). Thus, all four full-pipeline variants increase both review-error counts relative to Full.

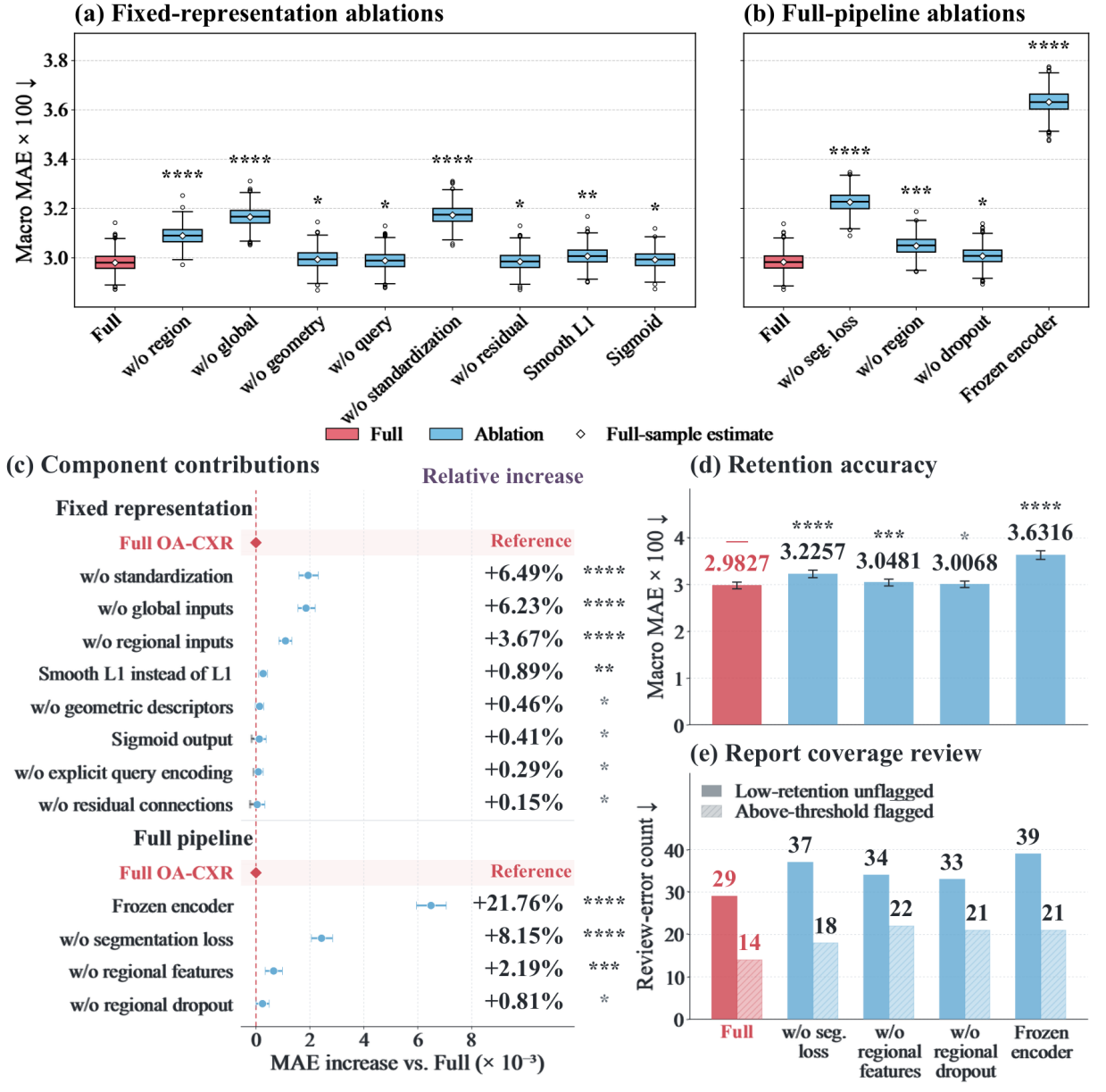


Figure 5: OA-CXR component ablations and their effects on retention accuracy and statement review. **(a,b)** Macro MAE distributions (×100) for fixed-representation/readout and full-pipeline ablations, respectively, from 1,000 paired grouped-bootstrap resamples. Boxes show interquartile ranges (IQRs); center lines, medians; whiskers, 1.5×IQR; circles, outliers; white diamonds, full-sample estimates. **(c)** Component-wise macro MAE increases ($\mathrm{MAE}_{\mathrm{variant}} - \mathrm{MAE}_{\mathrm{Full}}$, in 10^{-3} units), with paired 95% confidence intervals and relative percentage increases; positive differences favor Full. **(d)** Full-pipeline macro MAE (×100) with 95% grouped-bootstrap confidence intervals. **(e)** Fixed-threshold review errors among 784 eligible statements at threshold 0.9: solid bars show low-retention statements left unflagged ($y_q < 0.9$, $\hat{y}_q \geq 0.9$); hatched bars show above-threshold statements flagged ($y_q \geq 0.9$, $\hat{y}_q < 0.9$). Red denotes the separately retrained Full control for each ablation family; blue denotes ablations. All panels use development-selected checkpoints; macro MAE equally weights four public datasets. Adjusted significance: $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$; ns, $P \geq 0.05$.

Table 6
Auxiliary segmentation and statement-review results for full-pipeline ablations. Dice uses lung instances with nonempty reference masks; review errors cover 784 eligible statements at threshold 0.9. Full is the separately retrained full-pipeline control. Bold marks the highest Dice or lowest error count.

(a) Auxiliary segmentation performance

Condition	COVID-QU-Ex	Kermany/V7	NIH/CheXmask	Shenzhen
Without segmentation loss	0.1781	0.2311	0.1959	0.1956
Without regional features	**0.9276**	**0.9114**	**0.9359**	**0.9218**
Without regional-branch dropout	0.9186	0.9015	0.9279	0.9145
Frozen image encoder	0.8754	0.8614	0.8986	0.8828
OA-CXR (Full)	0.9218	0.9075	0.9309	0.9173

(b) Report-review errors

Condition	Low-retention unflagged	Above-threshold flagged
Without segmentation loss	37	18
Without regional features	34	22
Without regional-branch dropout	33	21
Frozen image encoder	39	21
OA-CXR (Full)	**29**	**14**

Table 7
Structural controls on 343,395 queries: source-specific and equally weighted macro MAE. OA-CXR is the selected main model; the geometry control shares its frozen segmentation model. Lower is better; bold marks the best result.

Method	COVID-QU-Ex	Kermany/V7	NIH/CheXmask	Shenzhen	Macro MAE
Global DenseNet	0.029773	0.039999	0.027359	0.032235	0.032341
Predicted-mask geometry + HGB	0.060557	0.067354	0.056755	0.064108	0.062193
OA-CXR (main model)	**0.026954**	**0.036829**	**0.025343**	**0.029397**	**0.029631**

Anatomical modeling improves retention accuracy over global features alone (Table 7). OA-CXR reduces macro MAE from 0.032341 for Global DenseNet to 0.029631 (8.38%), with source-specific paired intervals favoring OA-CXR on all four datasets. The predicted-mask geometry control, using 75 descriptors with HGB, yields macro MAE 0.062193. These controls support combining learned visual and anatomical features.

4.6. Review under Fixed Workloads

The fixed-workload analysis covers 6,656 MAIRA-2 reports: 445 contain 784 eligible statements, and 235 contain at least one finding-specific low-retention event. Review budgets apply to 445 eligible reports; recall is measured over 235 event-positive reports. We compare finding-specific OA-CXR, whole-lung-only OA-CXR, mean token log probability, and a retention–confidence combination calibrated on 761 eligible statements from 152 source groups in a separate partition. All methods score every eligible statement. Boundary ties receive equal fractional selection probabilities; intervals use 1,000 paired source-stratified grouped bootstrap resamples.

OA-CXR prioritizes event-positive reports (Table 8). Reviewing 50% of eligible reports captures 91.70% of event-positive reports (95% CI: 83.63–96.88%) versus 50.00% for mean token log probability, a gain of 41.70 percentage points (95% CI: 33.12–49.13). Among unreviewed eligible reports, low-retention event prevalence is 8.76% for OA-CXR versus 52.81% for mean token log probability.

Finding-specific query matching improves recall over whole-lung-only scores at the three larger budgets. At 50%, recall increases by 2.55 percentage points (95% CI: 0.22–3.73), supporting region-specific report ranking.

Table 8
Report-level recall (%) of low-retention events under fixed workloads. Budgets apply to 445 eligible reports; recall uses 235 finding-specific event-positive reports. Whole-lung-only scoring changes the ranking but retains finding-specific event labels. Higher is better; bold marks the highest recall per budget, including ties.

	Review budget				
Method	5%	10%	20%	30%	50%
Mean token log probability	4.2553	10.4255	20.8511	28.9362	50.0000
OA-CXR: whole-lung only	**9.4681**	**18.9362**	37.4468	55.9574	89.1489
OA-CXR: finding-specific	**9.4681**	**18.9362**	**37.8723**	**56.8085**	**91.7021**
OA-CXR + report confidence	**9.4681**	**18.9362**	**37.8723**	**56.8085**	**91.7021**

4.7. External Validation on an Independent Clinical Cohort

External evaluation uses 31,083 queries from 797 pediatric radiographs under the thirteen-presentation protocol and manual clinician references; 27.63% have low retention. Table 9 compares 14 configurations, and Fig. 6 shows ranking metrics, examples, and regional results.

OA-CXR achieves MAE 0.02810 (95% CI: 0.02721–0.02903), RMSE 0.05856, AUROC 0.97445 (95% CI: 0.97239–0.97641), AP 0.93931, and AURC 0.05055. Compared with Global DenseNet, it lowers MAE by 5.77% (0.02982 to 0.02810) and raises AUROC from 0.96929 to 0.97445; the paired MAE interval favors OA-CXR.

OA-CXR and TabM have similar overall MAE (0.02810 versus 0.02772), with a paired difference interval spanning zero. OA-CXR has higher AUROC (0.97445 versus 0.97266) and AP (0.93931 versus 0.93263), with paired intervals favoring OA-CXR; the AUROC difference remains significant after Holm adjustment (Fig. 6f; $P < 0.05$). By region, TabM has lower basal MAE, OA-CXR has lower inner-rim MAE, and whole-lung MAE shows no significant adjusted difference (Fig. 6g–l).

OA-CXR exceeds CheXpert-only, AutoQC CXR-NIQE, and BioViL-T on AUROC, AP, and AURC in the clinical cohort (Fig. 6(a–c)). Figure 6(d) shows six basal-retention examples around the 0.90 threshold with retention-readout input-gradient maps.

MAE increases to 0.08263 in the expanded 21-presentation panel with gray padding and placement variations. Supplementary Table S1 reports these results; the supplementary protocol covers cropped-only, photometric-perturbation, and prediction-consistency tests.

Figure 6 Clinical evaluation used 797 radiographs and 31,083 queries: thirteen presentations and three regions per source image. Manual clinician lung annotations defined reference retention.

a–c, ROC, precision–recall, and risk–coverage curves for reference retention below 0.90. In **c**, risk is low-retention prevalence after ranking by decreasing retention. Gray dashed lines show chance ROC or event prevalence. Curves cover 14 configurations; **e–l** compare 11 regressors.

d, Six post hoc examples of favorable OA-CXR performance: low retention (left) and above-threshold cases (right). Triplets show the source image, input, and input-gradient map. Hatching marks removed regions; cyan, retained basal reference anatomy; pink, absolute gradient magnitude. Paired arrows identify corresponding high-sensitivity locations. Each example lists reference retention, prediction, and absolute error.

e,f, Full-cohort MAE (**e**) and AUROC (**f**), with pointwise 95% percentile intervals.

g–l, Basal, inner-rim, and whole-lung MAE (**g–i**) and RMSE (**j–l**), scaled by 100. Box plots summarize 1,000 paired bootstrap resamples of 791 identity groups: boxes, IQR; lines, medians; whiskers, 1.5×IQR; open circles, outliers; white diamonds, full-sample estimates.

OA-CXR comparisons use approximate two-sided paired delete-one-group jackknife t-tests (790 degrees of freedom), Holm-adjusted across 80 comparisons. Significance: $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$; ns, $P \geq 0.05$. † favors the comparator; other significant differences favor OA-CXR. A dash denotes the OA-CXR reference.

4.8. Presentation Sensitivity and Reduced-Source Training

The perturbation challenge applies 21 conditions per source image: original presentation; single-edge crops at 5% and 30%; asymmetric opposing-edge crops at 7% and 23%; background-only cropping; brightness scaling at 0.7 and 1.3; Gaussian blur (radius 2); and padding changes. From 8,805 evaluation images, 184,864 valid presentations yield 554,592 queries. Forty-one padding conditions are inapplicable because the required padding or placement changes

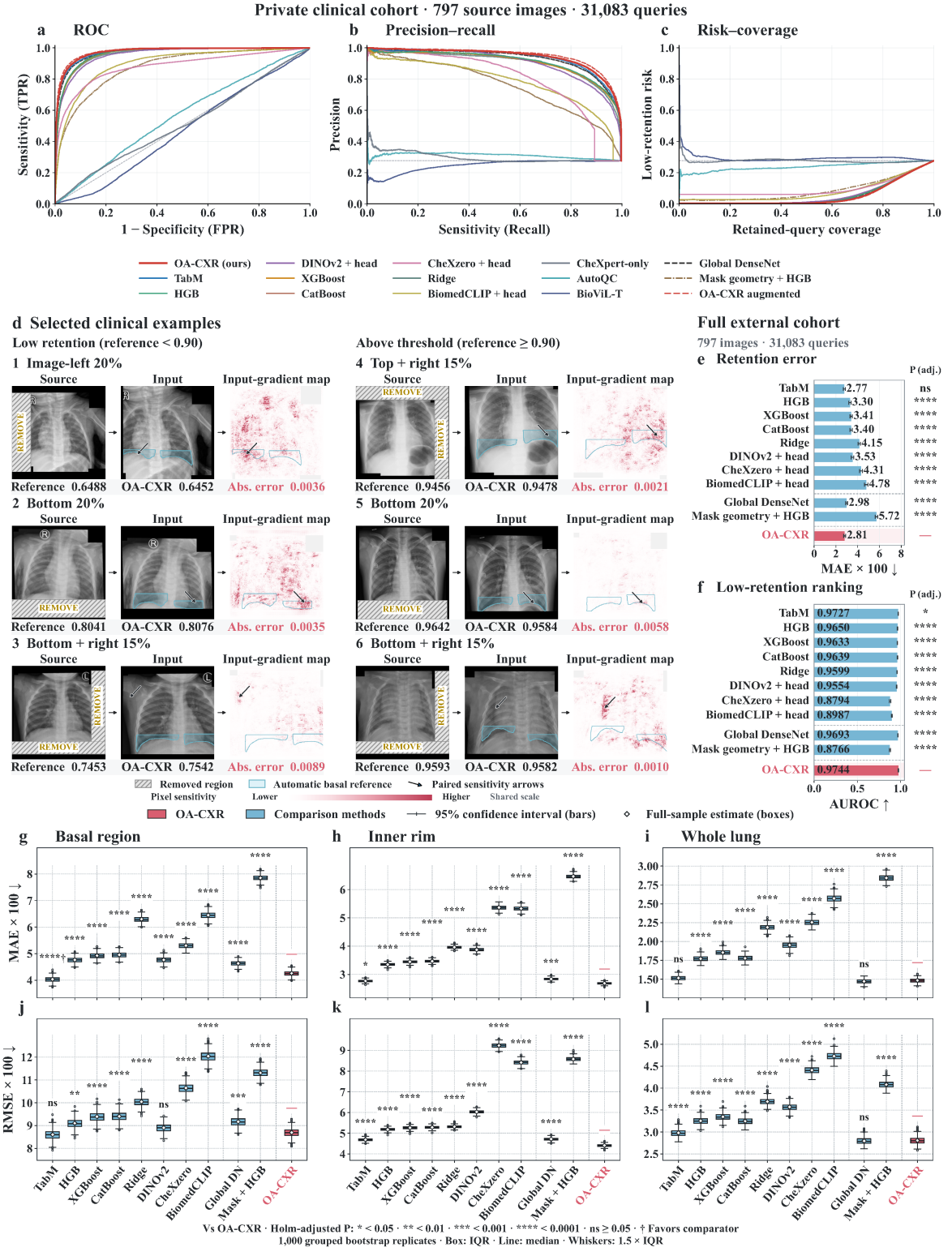


Figure 6: Clinical-cohort retention estimation and low-retention detection. Panel descriptions and statistical details follow on the next page.

Table 9
External validation against manual clinician references: 31,083 queries from 797 pediatric radiographs. Bold marks the best value per column. Image-based ranking signals have only AUROC, AP, and AURC. All 14 configurations have 100% scoring coverage.

Method	MAE↓	RMSE↓	AUROC↑	AP↑	AURC↓
HGB	0.03297	0.06332	0.96503	0.90897	0.05237
XGBoost	0.03409	0.06508	0.96327	0.90455	0.05299
CatBoost	0.03402	0.06507	0.96385	0.90427	0.05266
Ridge	0.04149	0.06898	0.95992	0.90013	0.05428
TabM	**0.02772**	0.05913	0.97266	0.93263	**0.05038**
DINOv2 + head	0.03532	0.06542	0.95540	0.90216	0.05678
CheXzero + head	0.04308	0.08526	0.87942	0.79770	0.09749
BiomedCLIP + head	0.04783	0.08913	0.89869	0.78253	0.08052
CheXpert-only	—	—	0.51024	0.29753	0.27489
AutoQC CXR-NIQE	—	—	0.55972	0.31052	0.23791
BioViL-T	—	—	0.46607	0.24584	0.29314
Global DenseNet	0.02982	0.06167	0.96929	0.92686	0.05201
Predicted-mask geometry + HGB	0.05719	0.08532	0.87662	0.75708	0.08743
OA-CXR	0.02810	**0.05856**	**0.97445**	**0.93931**	0.05055

Table 10
Perturbation sensitivity and reduced-source training. (a) Standard and challenge macro MAE, equally weighted across four sources; bold marks the lowest value. (b) Kermany/V7 results for the main and COVID-QU-Ex-only models; brackets give 95% intervals from 1,000 paired filename-family bootstrap resamples. Configurations differ in training sources, sample size, and optimizer updates.

(a) Standard and perturbation-challenge macro MAE

Method	Standard	Challenge
TabM	0.030767	0.087396
Global DenseNet	0.032341	0.087478
Predicted-mask geometry + HGB	0.062193	0.116153
OA-CXR: original training	**0.029631**	**0.077078**

(b) Reduced-source training evaluated on Kermany/V7

Training sources	MAE	RMSE	AUROC
COVID-QU-Ex + Kermany/V7	0.036829 [0.035414, 0.038341]	0.069322 [0.066670, 0.072131]	0.9705
COVID-QU-Ex only	0.074373 [0.071931, 0.077207]	0.127369 [0.123602, 0.131548]	0.8953

cannot be made. All four methods score every valid query. Transformations preserving the reference support leave the target unchanged.

The reduced-source model uses 16,192 COVID-QU-Ex fit images and 2,255 development images for checkpoint selection. Architecture, pretrained initialization, presentation schedule, and epoch counts match the main model; fitting and standardization use only COVID-QU-Ex. Evaluation covers 36,543 queries from 937 Kermany/V7 images across 360 filename families.

Table 10(b) compares MAE, RMSE, and AUROC, with paired filename-family bootstrap intervals for regression metrics. The training configurations differ in source composition, sample size, and optimizer-update count.

5. Discussion

Anatomical retention as a review signal. Source-relative anatomical retention provides an image-grounded basis for reviewing generated statements. The four-dataset analysis (Fig. 4) links prediction agreement and error uncertainty to event discrimination and risk–coverage. OA-CXR combines global and regional features to prioritize reports containing low-retention events; explicit statement-to-query mappings identify the anatomical basis for each flag.

Contributions of anatomical representation learning. Ablations (Fig. 5) show that encoder adaptation and segmentation supervision contribute most to full-pipeline accuracy, while regional inputs and standardization improve fixed-representation readout accuracy. The corresponding statement-review errors further demonstrate the downstream effects of these components. Structural and shared-feature controls further isolate the benefits of anatomical modeling and the residual readout. Together, the results support combining global context with localized anatomy.

Review allocation under fixed workloads. Under fixed review budgets, OA-CXR captures more event-positive reports and leaves fewer low-retention events among unreviewed eligible reports than generation confidence. Finding-specific scores further improve recall over whole-lung-only ranking. Anatomical retention therefore provides a direct criterion for allocating review effort to low-retention reports.

External validation in pediatric clinical imaging. The independent cohort covers pediatric radiographs from ages 10 days to 13 years, including naturally incomplete acquisitions. Results against manual clinician references demonstrate transfer without model adaptation. OA-CXR and TabM have similar overall MAE, but OA-CXR yields higher AUROC and AP for low-retention ranking. Clinician-assessed report-review outcomes extend the evaluation beyond retention estimates.

Presentation sensitivity and future directions. Controlled public-data presentations and the independent clinical cohort, with naturally incomplete acquisitions, manual references, and clinician-assessed review outcomes, provide complementary evaluations. Report-review findings concern parser-defined statements from one generator; further work can test additional generators and report content. Patient-level partitioning and cross-source deduplication would strengthen generalization assessment.

6. Conclusion

OA-CXR guides selective chest radiograph report review using source-relative anatomical retention. A two-stage global–regional estimator predicts retention from the presented image and an anatomical query; statement mappings convert scores into flags and report priorities. OA-CXR has the lowest MAE among nine retention regressors across four datasets. Reviewing 50% of eligible reports captures 91.7% of event-positive reports versus 50.0% using generation confidence. On 797 pediatric radiographs with manual clinician references, the frozen model achieves MAE 0.02810 and AUROC 0.97445. Anatomical retention supports prioritizing reports with low-retention events under fixed review budgets.

Data availability

The four public datasets are available from their respective sources; the project repository linked in the abstract provides download links and access instructions.

The independent pediatric clinical cohort from Shenzhen University of Advanced Technology General Hospital is not publicly available due to patient privacy, confidentiality, and institutional data protection requirements.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Angelopoulos, A.N., Bates, S., Fisch, A., Lei, L., Schuster, T., 2024. Conformal risk control, in: International Conference on Learning Representations. URL: https://openreview.net/forum?id=33XGfHLtZg.

[2] Bannur, S., et al., 2023. Learning to exploit temporal structure for biomedical vision-language processing, in: Proc. IEEE/CVF CVPR, pp. 15016–15027. URL: https://openaccess.thecvf.com/content/CVPR2023/html/Bannur_Learning_To_Exploit_Temporal_Structure_for_Biomedical_Vision-Language_Processing_CVPR_2023_paper.html.

[3] Bannur, S., et al., 2024. MAIRA-2: Grounded radiology report generation. arXiv:2406.04449. URL: https://arxiv.org/abs/2406.04449, doi:10.48550/arXiv.2406.04449.

[4] Boecking, B., Usuyama, N., Bannur, S., Castro, D.C., Schwaighofer, A., Hyland, S., Wetscherek, M., Naumann, T., Nori, A., Alvarez-Valle, J., Poon, H., Oktay, O., 2022. Making the most of text semantics to improve biomedical vision–language processing, in: Computer Vision – ECCV 2022, pp. 1–21. URL: https://arxiv.org/abs/2204.09817, doi:10.1007/978-3-031-20059-5_1.

[5] Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting system, in: Proc. ACM SIGKDD, pp. 785–794. URL: https://doi.org/10.1145/2939672.2939785, doi:10.1145/2939672.2939785.

[6] Chen, Z., Shen, Y., Song, Y., Wan, X., 2021. Cross-modal memory networks for radiology report generation, in: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), pp. 5904–5914. URL: https://aclanthology.org/2021.acl-long.459/, doi:10.18653/v1/2021.acl-long.459.

[7] Chen, Z., Song, Y., Chang, T.H., Wan, X., 2020. Generating radiology reports via memory-driven transformer, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing. URL: https://arxiv.org/abs/2010.16056, doi:10.18653/v1/2020.emnlp-main.112.

[8] Chen, Z., et al., 2024. CheXagent: Towards a foundation model for chest X-Ray interpretation. arXiv:2401.12208v1. URL: https://arxiv.org/abs/2401.12208v1, doi:10.48550/arXiv.2401.12208.

[9] Cohen, J.P., et al., 2022. TorchXRayVision: A library of chest X-ray datasets and models, in: Proc. Medical Imaging with Deep Learning, pp. 231–249. URL: https://proceedings.mlr.press/v172/cohen22a.html.

[10] Corbière, C., Thome, N., Bar-Hen, A., Cord, M., Pérez, P., 2019. Addressing failure prediction by learning model confidence, in: Advances in Neural Information Processing Systems. URL: https://arxiv.org/abs/1910.04851.

[11] D'Amour, A., et al., 2022. Underspecification presents challenges for credibility in modern machine learning. Journal of Machine Learning Research 23, 1–61. URL: https://www.jmlr.org/beta/papers/v23/20-1335.html.

[12] DeGrave, A.J., Janizek, J.D., Lee, S.I., 2021. AI for radiographic COVID-19 detection selects shortcuts over signal. Nature Machine Intelligence 3, 610–619. URL: https://www.nature.com/articles/s42256-021-00338-7, doi:10.1038/s42256-021-00338-7.

[13] Delbrouck, J.B., Chambon, P., Bluethgen, C., Tsai, E., Almusa, O., Langlotz, C.P., 2022. Improving the factual correctness of radiology report generation with semantic rewards, in: Findings of the Association for Computational Linguistics: EMNLP 2022. URL: https://arxiv.org/abs/2210.12186.

[14] El-Yaniv, R., Wiener, Y., 2010. On the foundations of noise-free selective classification. Journal of Machine Learning Research 11, 1605–1641. URL: https://www.jmlr.org/papers/v11/el-yaniv10a.html.

[15] Friedman, J.H., 2001. Greedy function approximation: A gradient boosting machine. The Annals of Statistics 29, 1189–1232. URL: https://doi.org/10.1214/aos/1013203451, doi:10.1214/aos/1013203451.

[16] Gaggion, N., Mansilla, L., Mosquera, C., Milone, D.H., Ferrante, E., 2023. Improving anatomical plausibility in medical image segmentation via hybrid graph neural networks: Applications to chest X-Ray analysis. IEEE Transactions on Medical Imaging 42, 546–556. URL: https://doi.org/10.1109/TMI.2022.3224660, doi:10.1109/TMI.2022.3224660.

[17] Gaggion, N., Mosquera, C., Aineseder, M., Mansilla, L., Milone, D., Ferrante, E., 2025. CheXmask Database: A large-scale dataset of anatomical segmentation masks for chest X-ray images. PhysioNet URL: https://physionet.org/content/chexmask-cxr-segmentation-data/1.0.0/, doi:10.13026/3705-zg36. version 1.0.0.

[18] Gaggion, N., Mosquera, C., Mansilla, L., Saidman, J.M., Aineseder, M., Milone, D.H., Ferrante, E., 2024. CheXmask: A large-scale dataset of anatomical segmentation masks for multi-center chest X-ray images. Scientific Data 11, 511. URL: https://www.nature.com/articles/s41597-024-03358-1, doi:10.1038/s41597-024-03358-1.

[19] Gal, Y., Ghahramani, Z., 2016. Dropout as a bayesian approximation: Representing model uncertainty in deep learning, in: Proceedings of the 33rd International Conference on Machine Learning, pp. 1050–1059. URL: https://proceedings.mlr.press/v48/gal16.html.

[20] Geifman, Y., El-Yaniv, R., 2017. Selective classification for deep neural networks, in: Advances in Neural Information Processing Systems. URL: https://proceedings.neurips.cc/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html.

[21] Geifman, Y., El-Yaniv, R., 2019. SelectiveNet: A deep neural network with an integrated reject option, in: Proceedings of the 36th International Conference on Machine Learning, pp. 2151–2159. URL: https://proceedings.mlr.press/v97/geifman19a.html.

[22] Geirhos, R., Jacobsen, J.H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., Wichmann, F.A., 2020. Shortcut learning in deep neural networks. Nature Machine Intelligence 2, 665–673. URL: https://www.nature.com/articles/s42256-020-00257-z, doi:10.1038/s42256-020-00257-z.

[23] Gorishniy, Y., Kotelnikov, A., Babenko, A., 2025. TabM: Advancing tabular deep learning with parameter-efficient ensembling, in: Proc. ICLR. URL: https://proceedings.iclr.cc/paper_files/paper/2025/hash/c1ba41c694834aeef91ae161711d4939-Abstract-Conference.html.

[24] Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q., 2017. On calibration of modern neural networks, in: Proceedings of the 34th International Conference on Machine Learning, pp. 1321–1330. URL: https://proceedings.mlr.press/v70/guo17a.html.

[25] Hendrycks, D., Dietterich, T., 2019. Benchmarking neural network robustness to common corruptions and perturbations, in: International Conference on Learning Representations. URL: https://arxiv.org/abs/1903.12261.

[26] Hendrycks, D., Gimpel, K., 2017. A baseline for detecting misclassified and out-of-distribution examples in neural networks, in: International Conference on Learning Representations. URL: https://arxiv.org/abs/1610.02136.

[27] Hoerl, A.E., Kennard, R.W., 1970. Ridge regression: Biased estimation for nonorthogonal problems. Technometrics 12, 55–67. URL: https://doi.org/10.1080/00401706.1970.10488634, doi:10.1080/00401706.1970.10488634.

[28] Huang, G., Liu, Z., van der Maaten, L., Weinberger, K.Q., 2017. Densely connected convolutional networks, in: Proc. IEEE CVPR, pp. 4700–4708. URL: https://openaccess.thecvf.com/content_cvpr_2017/html/Huang_Densely_Connected_Convolutional_CVPR_2017_paper.html.

[29] Huang, S.C., Shen, L., Lungren, M.P., Yeung, S., 2021. GLoRIA: A multimodal global-local representation learning framework for label-efficient medical image recognition, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 3942–3951. URL: https://ieeexplore.ieee.org/document/9710099.

[30] Irvin, J., et al., 2019. CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison, in: Proc. AAAI, pp. 590–597. URL: https://ojs.aaai.org/index.php/AAAI/article/view/3834, doi:10.1609/aaai.v33i01.3301590.

[31] Jaeger, S., et al., 2014. Two public chest X-ray datasets for computer-aided screening of pulmonary diseases. Quantitative Imaging in Medicine and Surgery 4, 475–477. URL: https://qims.amegroups.org/article/view/5132/html, doi:10.3978/j.issn.2223-4292.2014.11.20.

[32] Jain, S., et al., 2021. RadGraph: Extracting clinical entities and relations from radiology reports, in: Proc. NeurIPS Datasets and Benchmarks. URL: https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/c8ffe9a587b126f152ed3d89a146b445-Abstract-round1.html.

[33] Jiang, H., Kim, B., Guan, M.Y., Gupta, M., 2018. To trust or not to trust a classifier, in: Advances in Neural Information Processing Systems. URL: https://arxiv.org/abs/1805.11783.

[34] Jing, B., Xie, P., Xing, E., 2018. On the automatic generation of medical imaging reports, in: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 2577–2586. URL: https://aclanthology.org/P18-1240/, doi:10.18653/v1/P18-1240.

[35] Johnson, A.E.W., Pollard, T.J., Berkowitz, S.J., Greenbaum, N.R., Lungren, M.P., Deng, C.y., Mark, R.G., Horng, S., 2019. MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports. Scientific Data 6, 317. URL: https://www.nature.com/articles/s41597-019-0322-0, doi:10.1038/s41597-019-0322-0.

[36] Kermany, D., Zhang, K., Goldbaum, M., 2018. Labeled optical coherence tomography (OCT) and chest X-Ray images for classification. Mendeley Data, version 2. URL: https://data.mendeley.com/datasets/rscbjbr9sj/2, doi:10.17632/rscbjbr9sj.2.

[37] Koh, P.W., et al., 2021. WILDS: A benchmark of in-the-wild distribution shifts, in: Proceedings of the 38th International Conference on Machine Learning, pp. 5637–5664. URL: https://proceedings.mlr.press/v139/koh21a.html.

[38] Lakshminarayanan, B., Pritzel, A., Blundell, C., 2017. Simple and scalable predictive uncertainty estimation using deep ensembles, in: Advances in Neural Information Processing Systems. URL: https://arxiv.org/abs/1612.01474.

[39] Li, Y., Liang, X., Hu, Z., Xing, E.P., 2018. Hybrid retrieval-generation reinforced agent for medical image report generation, in: Advances in Neural Information Processing Systems. URL: https://papers.neurips.cc/paper_files/paper/2018/hash/e07413354875be01a996dc560274708e-Abstract.html.

[40] Liu, G., Hsu, T.M.H., McDermott, M., Boag, W., Weng, W.H., Szolovits, P., Ghassemi, M., 2019. Clinically accurate chest X-Ray report generation, in: Proceedings of the 4th Machine Learning for Healthcare Conference, pp. 249–269. URL: https://proceedings.mlr.press/v106/liu19a.html.

[41] Madras, D., Pitassi, T., Zemel, R., 2018. Predict responsibly: Improving fairness and accuracy by learning to defer, in: Advances in Neural Information Processing Systems. URL: https://papers.nips.cc/paper_files/paper/2018/hash/09d37c08f7b129e96277388757530c72-Abstract.html.

[42] Mittal, A., Soundararajan, R., Bovik, A.C., 2013. Making a "completely blind" image quality analyzer. IEEE Signal Processing Letters 20, 209–212. URL: https://doi.org/10.1109/LSP.2012.2227726, doi:10.1109/LSP.2012.2227726.

[43] Mozannar, H., Sontag, D., 2020. Consistent estimators for learning to defer to an expert, in: Proceedings of the 37th International Conference on Machine Learning, pp. 7076–7087. URL: https://proceedings.mlr.press/v119/mozannar20b.html.

[44] Oakden-Rayner, L., Dunnmon, J., Carneiro, G., Ré, C., 2020. Hidden stratification causes clinically meaningful failures in machine learning for medical imaging, in: Proceedings of the ACM Conference on Health, Inference, and Learning. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC7665161/, doi:10.1145/3368555.3384468.

[45] Oquab, M., et al., 2024. DINOv2: Learning robust visual features without supervision. Transactions on Machine Learning Research URL: https://openreview.net/forum?id=a68SUt6zFt.

[46] Ostmeier, S., Xu, J., Chen, Z., Varma, M., Blankemeier, L., Bluethgen, C., Michalson, A.E., Moseley, M., Langlotz, C., Chaudhari, A.S., Delbrouck, J.B., 2024. GREEN: Generative radiology report evaluation and error notation, in: Findings of the Association for Computational Linguistics: EMNLP 2024, pp. 374–390. URL: https://aclanthology.org/2024.findings-emnlp.21/, doi:10.18653/v1/2024.findings-emnlp.21.

[47] Ovadia, Y., Fertig, E., Ren, J., Nado, Z., Sculley, D., Nowozin, S., Dillon, J.V., Lakshminarayanan, B., Snoek, J., 2019. Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift, in: Advances in Neural Information Processing Systems. URL: https://arxiv.org/abs/1906.02530.

[48] Pedregosa, F., et al., 2011. Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12, 2825–2830. URL: https://jmlr.org/papers/v12/pedregosa11a.html.

[49] Pellegrini, C., Özsoy, E., Busam, B., Navab, N., Keicher, M., 2025. RaDialog: A large vision-language model for radiology report generation and conversational assistance, in: Medical Imaging with Deep Learning. URL: https://arxiv.org/abs/2311.18681v3.

[50] Pollard, T., Moody, B.E., Lehman, L.w.H., Gow, B.J., Fernandes, C., Xie, C., Johnson, A., Mark, R.G., Heldt, T., 2026. PhysioNet as a global platform for biomedical research. Nature Health 1, 792–795. URL: https://doi.org/10.1038/s44360-026-00096-z, doi:10.1038/s44360-026-00096-z.

[51] Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V., Gulin, A., 2018. CatBoost: Unbiased boosting with categorical features, in: Advances in Neural Information Processing Systems. URL: https://proceedings.neurips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html.

[52] Raghu, M., Blumer, K., Sayres, R., Obermeyer, Z., Kleinberg, B., Mullainathan, S., Kleinberg, J., 2019. Direct uncertainty prediction for medical second opinions, in: Proceedings of the 36th International Conference on Machine Learning, pp. 5281–5290. URL: https://proceedings.mlr.press/v97/raghu19a.html.

[53] Selby, I.A., et al., 2025. A pipeline for automated quality control of chest radiographs. Radiology: Artificial Intelligence 7, e240003. URL: https://pubs.rsna.org/doi/10.1148/ryai.240003, doi:10.1148/ryai.240003.

[54] Stirenko, S., et al., 2018. Chest X-Ray analysis of tuberculosis by deep learning with segmentation and augmentation, in: Proc. IEEE ELNANO, pp. 422–428. URL: https://arxiv.org/abs/1803.01199, doi:10.1109/ELNANO.2018.8477564.

[55] Tahir, A.M., Chowdhury, M.E.H., Qiblawey, Y., Khandakar, A., Rahman, T., Kiranyaz, S., Khurshid, U., Ibtehaz, N., Mahmud, S., Ezeddin, M., 2021a. COVID-QU-Ex. Kaggle, version 7. URL: https://www.kaggle.com/dsv/3122958, doi:10.34740/kaggle/dsv/3122958.

[56] Tahir, A.M., et al., 2021b. COVID-19 infection localization and severity grading from chest X-ray images. Computers in Biology and Medicine 139, 105002. URL: https://doi.org/10.1016/j.compbiomed.2021.105002, doi:10.1016/j.compbiomed.2021.105002.

[57] Tanida, T., Müller, P., Kaissis, G., Rueckert, D., 2023. Interactive and explainable region-guided radiology report generation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 7433–7442. URL: https://arxiv.org/abs/2304.08295, doi:10.1109/CVPR52729.2023.00718.

[58] Tiu, E., Talius, E., Patel, P., Langlotz, C.P., Ng, A.Y., Rajpurkar, P., 2022. Expert-level detection of pathologies from unannotated chest X-ray images via self-supervised learning. Nature Biomedical Engineering 6, 1399–1406. URL: https://www.nature.com/articles/s41551-022-00936-9, doi:10.1038/s41551-022-00936-9.

[59] V7 Labs, . COVID-19 X-ray dataset. GitHub repository. URL: https://github.com/v7labs/covid-19-xray-dataset. accessed: Sep. 25, 2026; publication year not specified.

[60] Wang, C., Zhou, W., Ghosh, S., Batmanghelich, K., Li, W., 2025. Semantic consistency-based uncertainty quantification for factuality in radiology report generation, in: Findings of the Association for Computational Linguistics: NAACL 2025, pp. 1739–1754. URL: https://aclanthology.org/2025.findings-naacl.95/, doi:10.18653/v1/2025.findings-naacl.95.

[61] Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., Summers, R.M., 2017. ChestX-ray8: Hospital-scale chest X-Ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases, in: Proc. IEEE CVPR, pp. 2097–2106. URL: https://openaccess.thecvf.com/content_cvpr_2017/html/Wang_ChestX-ray8_Hospital-Scale_Chest_CVPR_2017_paper.html.

[62] Wilder, B., Horvitz, E., Kamar, E., 2020. Learning to complement humans, in: Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, pp. 1526–1533. URL: https://www.ijcai.org/proceedings/2020/212, doi:10.24963/ijcai.2020/212.

[63] Wu, J.T., Agu, N.N., Lourentzou, I., Sharma, A., Paguio, J.A., Yao, J.S., Dee, E.C., Mitchell, W., Kashyap, S., Giovannini, A., Celi, L.A., Moradi, M., 2021. Chest ImaGenome dataset for clinical reasoning, in: Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks. URL: https://openreview.net/forum?id=H-d5634yVi.

[64] Yu, F., Endo, M., Krishnan, R., Pan, I., Tsai, A., Reis, E.P., Fonseca, E.K.U.N., Lee, H.M.H., Abad, Z.S.H., Ng, A.Y., Langlotz, C.P., Venugopal, V.K., Rajpurkar, P., 2023. Evaluating progress in automatic chest X-ray radiology report generation. Patterns 4, 100802. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC10499844/, doi:10.1016/j.patter.2023.100802.

[65] Zech, J.R., Badgeley, M.A., Liu, M., Costa, A.B., Titano, J.J., Oermann, E.K., 2018. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. PLOS Medicine 15, e1002683. URL: https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.1002683, doi:10.1371/journal.pmed.1002683.

[66] Zhang, S., et al., 2023. BiomedCLIP: A multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. arXiv:2303.00915. URL: https://arxiv.org/abs/2303.00915, doi:10.48550/arXiv.2303.00915.

[67] Zhang, Y., Jiang, H., Miura, Y., Manning, C.D., Langlotz, C.P., 2022. Contrastive learning of medical visual representations from paired images and text, in: Proceedings of the 7th Machine Learning for Healthcare Conference, pp. 2–25. URL: https://proceedings.mlr.press/v182/zhang22a.html.